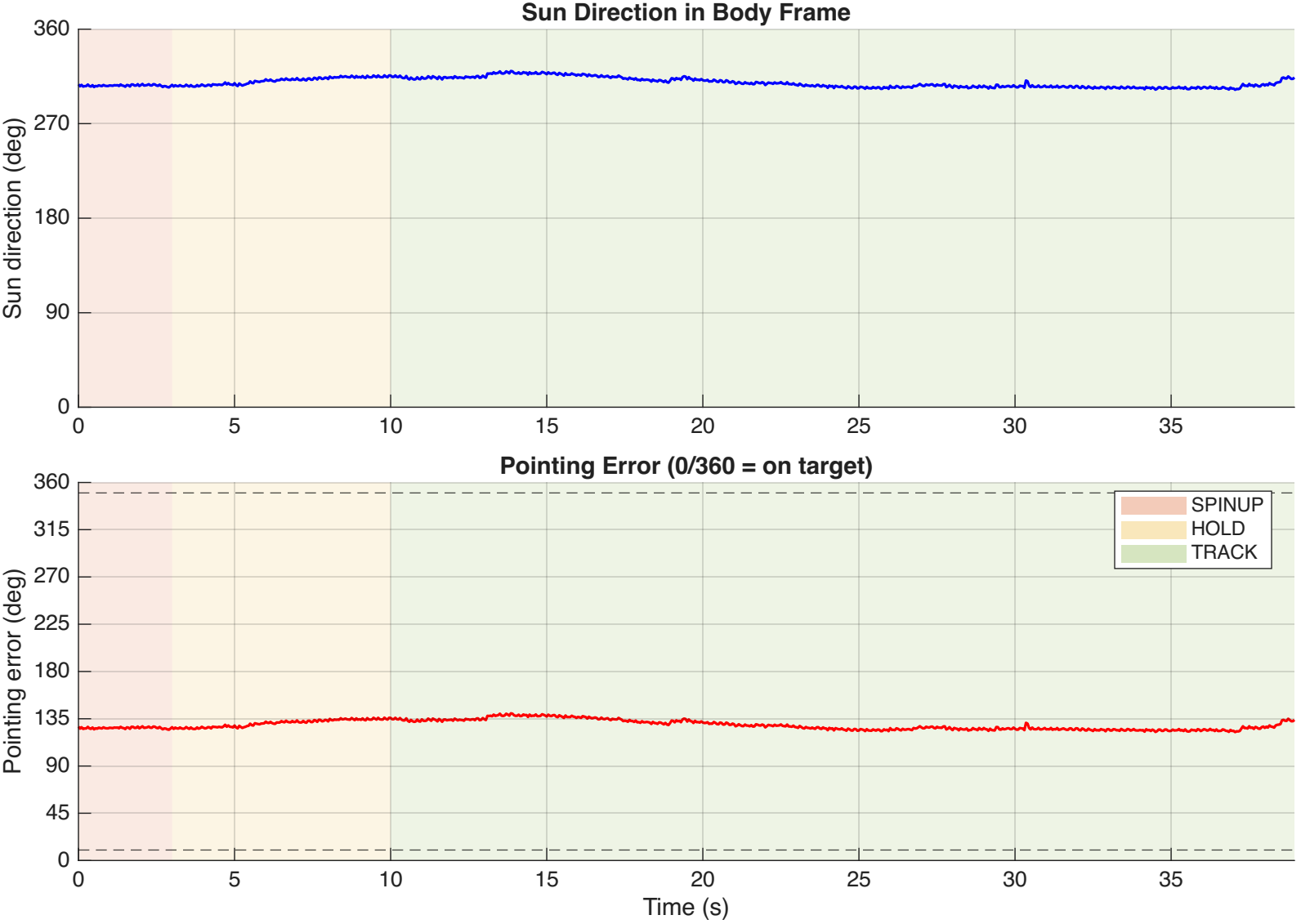
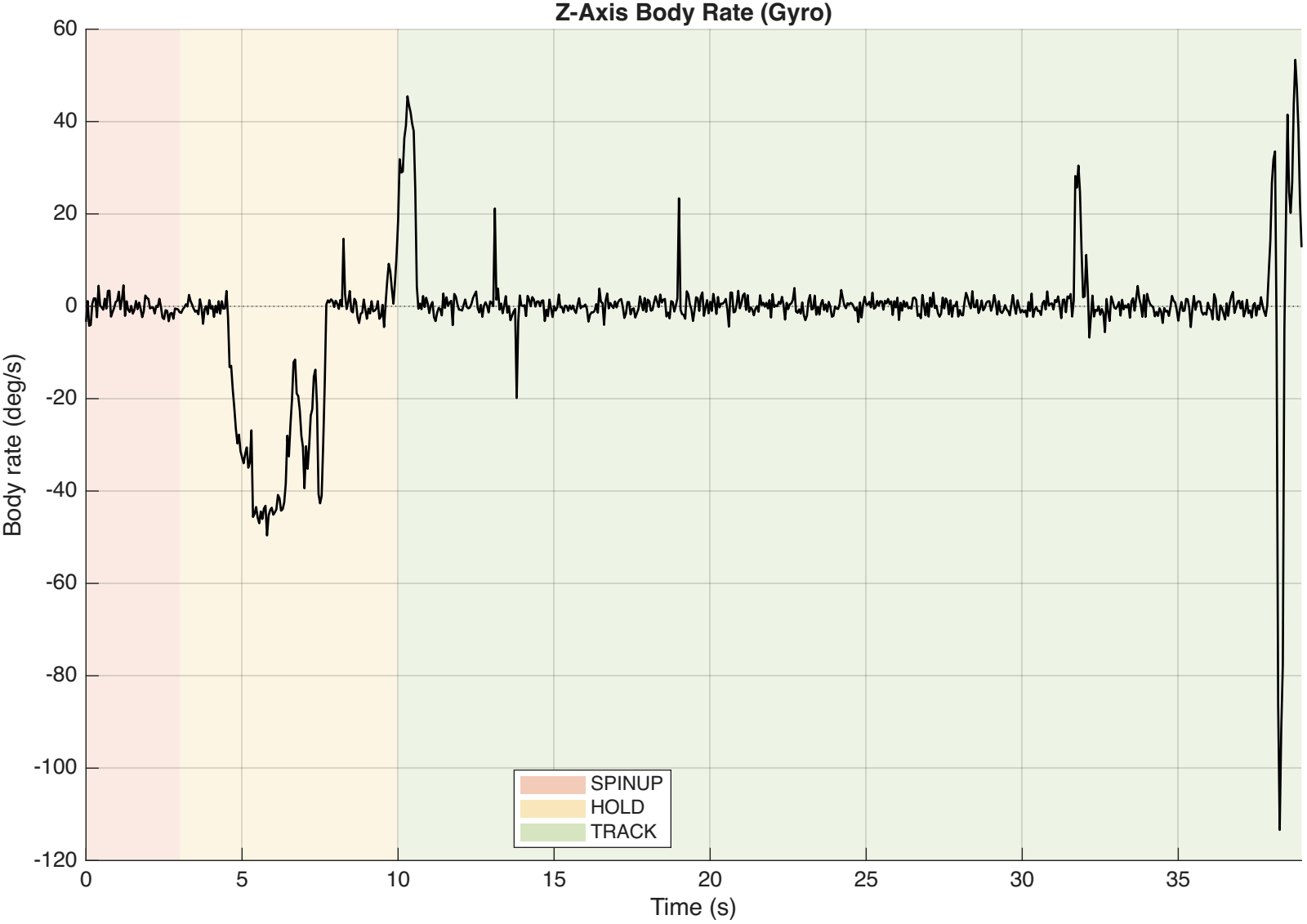
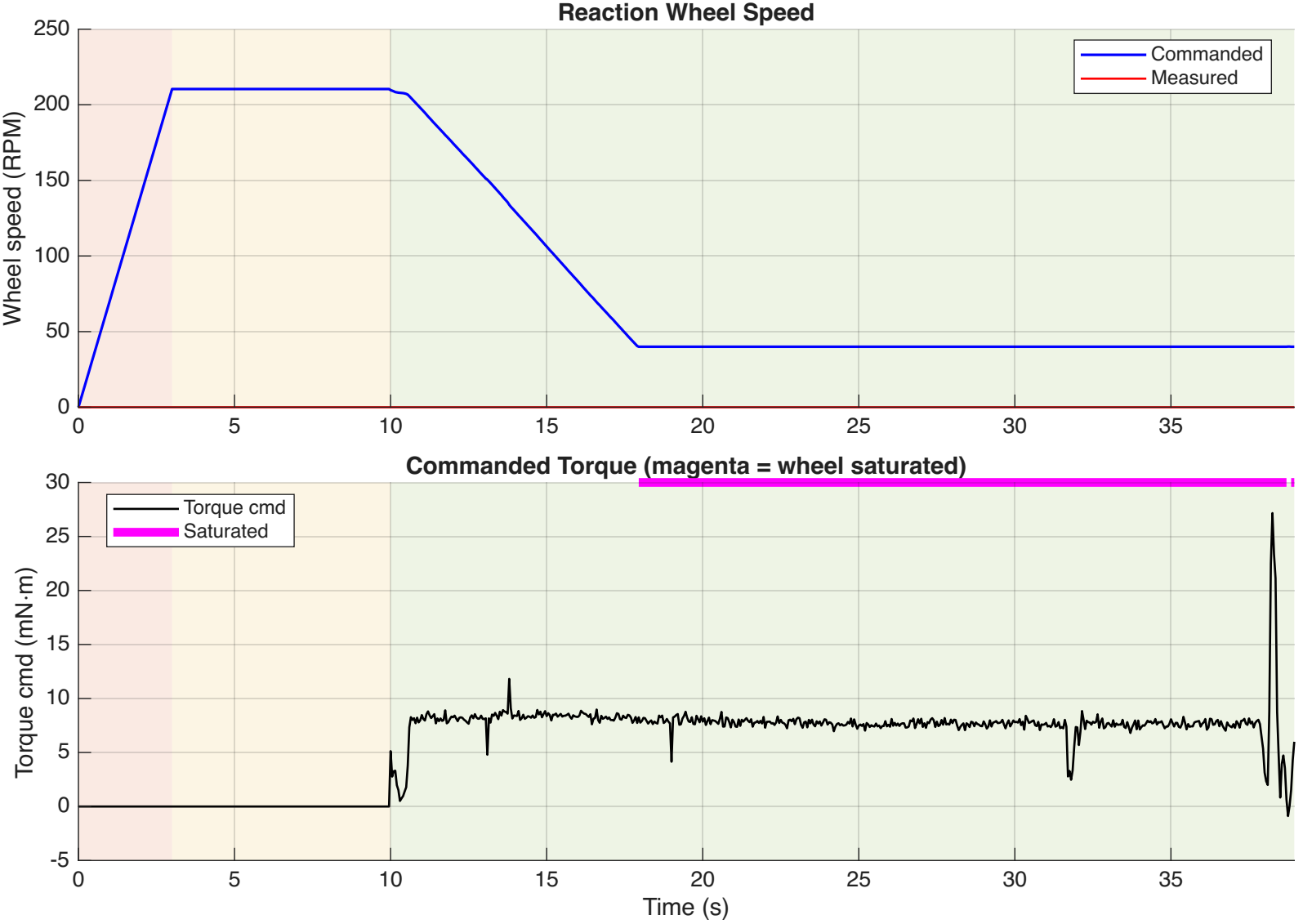
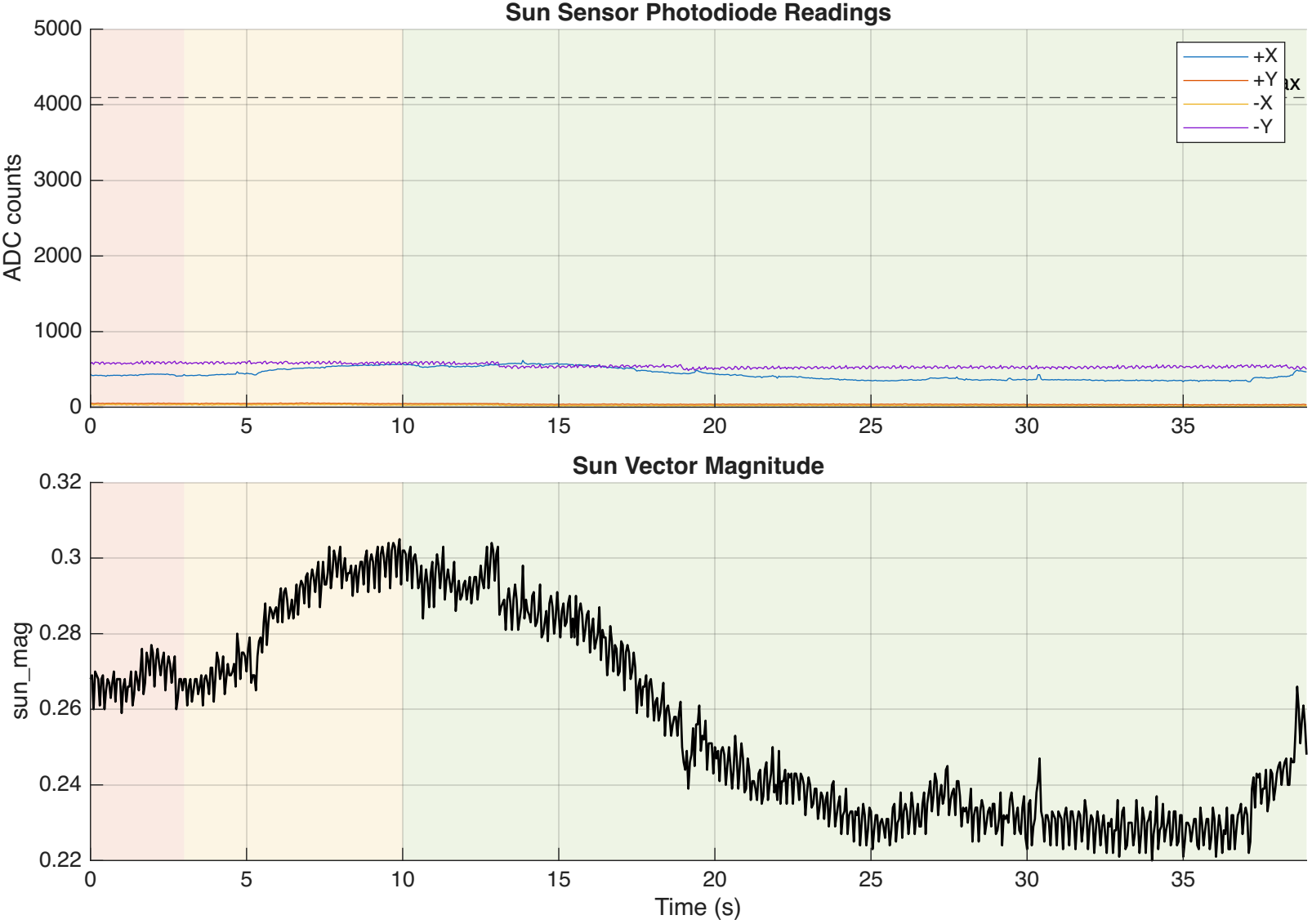
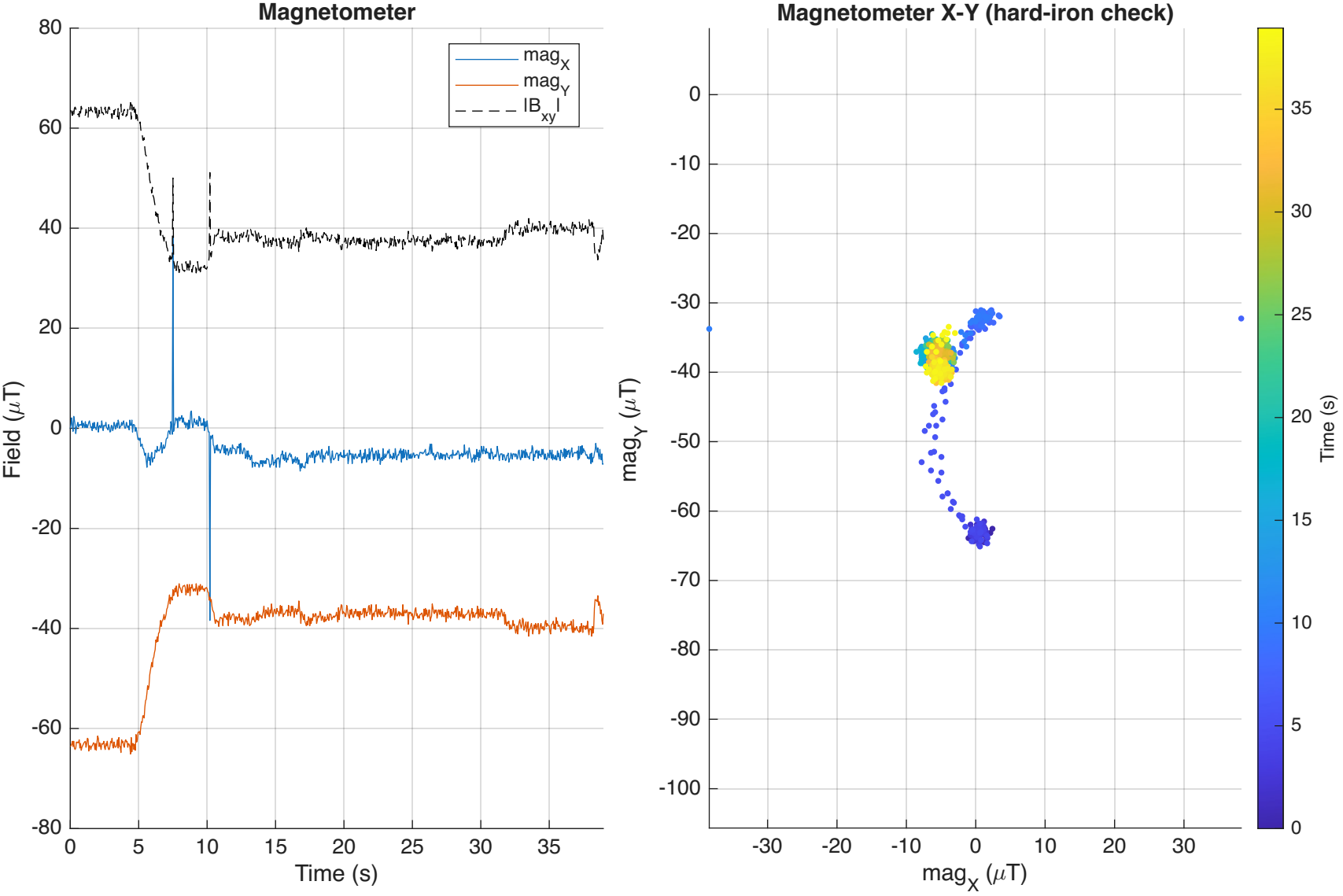
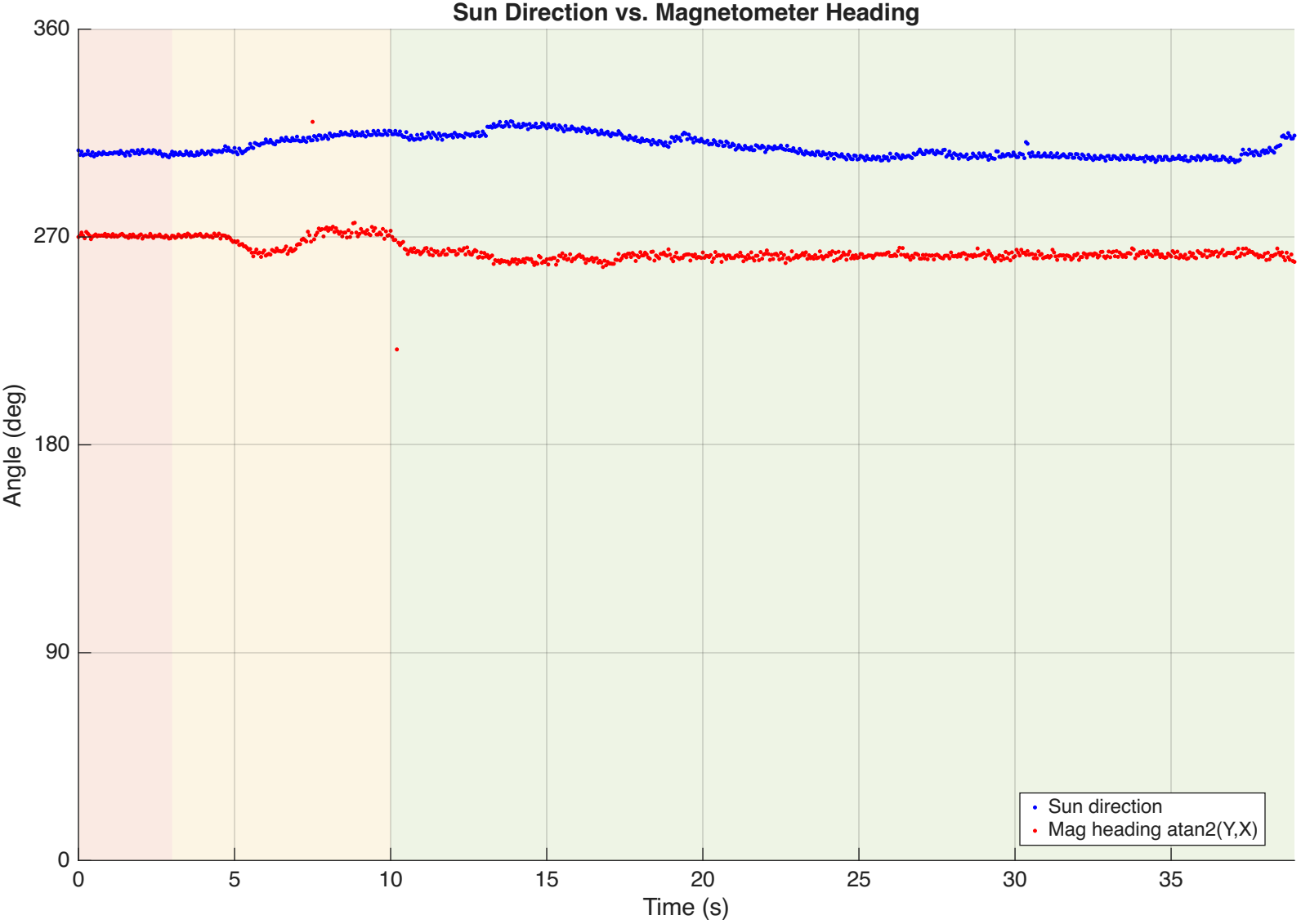

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
5
10
15
20
25
30
35
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)


Z-Axis Body Rate (Gyro)
Body rate (deg/s)
60
40
20
0
-20
-40
-60
-80
-100
-120
0
5
10
15
20
25
30
35
Time (s)
SPINUP
HOLD
TRACK


Reaction Wheel Speed
Commanded
Measured
Wheel speed (RPM)
250
200
150
100
50
0
0
5
10
15
20
25
30
35
Commanded Torque (magenta = wheel saturated)
Torque cmd
Saturated
Torque cmd (mN·m)
30
25
20
15
10
5
0
-5
0
5
10
15
20
25
30
35
Time (s)


Sun Sensor Photodiode Readings
ADC counts
5000
4000
3000
2000
1000
0
+X
+Y
-X
-Y
0
5
10
15
20
25
30
35
Sun Vector Magnitude
sun_mag
0.32
0.3
0.28
0.26
0.24
0.22
Time (s)


Magnetometer
magX
magY
|Bxy|
Field (μT)
Time (s)
Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)


Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
5
10
15
20
25
30
35
Time (s)
Sun direction
Mag heading atan2(Y,X)